Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms a correct equation by applying conservation of energy with KE and PE − substituting the values given	AO3.4	M1	KE gained = PE lost $\frac{1}{2}mv^2 = mgh$ $\frac{1}{2}(75)v^2 = 75g(25)$ $v^2 = 50g$ Or $v^2 = 490$ $v = 22$ m s^{-1} (2sf)
	Solves the equation correctly to obtain the value of v − must be rounded correctly to 2 significant figures	AO3.2a	A1	
(b)	Forms expressions for EPE and PE	AO3.1b	M1	PE lost = EPE gained $\frac{\lambda x^2}{2l} = mgh$ $\frac{3200x^2}{2(25)} = 75(9.8)(25+x)$ $64x^2 - 735x - 18375 = 0$ $x = 23.6\ldots$ 23.6 + 25 = 48.6 m 48.6 m < 50 m Dominic does not get wet
	Obtains fully correct expressions	AO1.1b	A1	
	Solves a three-term quadratic equation or substitutes correct distances for a total length of 50 m into both expressions.	AO1.1a	M1	
	Obtains the correct maximum extension of the cord or obtains the correct values for the two energies	AO1.1b	A1	
	Compares 'their' results and concludes that Dominic does not get wet	AO3.2a	E1F	
(c)	Comments that Dominic has size and considers the effect this might have on the distance fallen	AO3.2b	E1	Dominic has size so distance descended would be greater than 48.6 m He might get wet if he was taller than 1.4 m
	States clearly whether Dominic does or does not get wet, justifying their conclusion	AO3.5a	E1F	
				Total 9 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms an energy equation between PE and EPE	AO3.3	M1	$0.5 \times 2 \times 10 = \frac{\lambda}{2 \times 0.2} \times 0.4^2$
	Obtains correct value for λ or k	AO1.1b	A1	
	Using Hooke's Law to find	AO3.4	M1	

	extension at equilibrium			$\lambda = \frac{10}{0.4}$ = 25 Nm^{-1}
	Obtains correct extension at equilibrium using 'their' λ or k	AO1.1b	A1F	
	Forms equation using conservation of energy.	AO3.4	M1	$2 \times 10 = \frac{25}{0.2}e$
	Obtains the correct speed. (Only accept V = 2 m s^{-1} or V = 2.0 m s^{-1})	AO1.1b	A1	$e = \frac{2 \times 10 \times 0.2}{25}$ = 0.16 $2 \times 10 \times 0.26 = \frac{1}{2} \times 2v^2 + \frac{25 \times 0.16^2}{2 \times 0.4}$ v^2 = 3.6 v = 1.897… = 2.0 m s^{-1}(to 1sf)
(b)	States appropriate refinement.	AO3.5c	E1	Could include air resistance.
				Total 7 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates correct EPE.	AO1.1b	B1	$EPE = \frac{65 \times 0.7^2}{2 \times 1.3} = 12.25J$
	Finds correct value for friction.	AO3.1b	B1	Friction 2 × 9.8 × cos(30°) × 0.6 = 10.184 N
	Forms work-energy equation, with at least two correct terms.	AO3.3	M1	EPE = $F_r(2 - d)$ + $mg(2 - d)\sin(30°)$ + $\frac{65(d-1.3)^2}{2 \times 1.3}$
	Obtains work-energy equations with correct terms and correct signs.	AO1.1b	A1	
	Solves 'their' energy equation to obtain two solutions. (PI)	AO1.1a	M1	12.25 = 10.184(2 – d) + 9.8(2 – d) + 25(d – 1.3)^2
	Rejects d = 2 with a reason.	AO2.4	A1	
	Completes rigorous argument to obtain 1.4.	AO2.1	R1	$25d^2 - 84.984d + 69.968 = 0$

	AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips. All energy terms must be included and two solutions stated for the quadratic equation.			d = 1.39936 or 2 d = 2 represents the starting position so is not a solution given that the particle moves d = 1.4 m
(b)	Calculates the component of weight parallel to the plane.	AO3.1b	M1	Component of weight down plane = 2×9.8sin(30°) = 9.6 Limiting value of friction = 10.184 > 9.6 (and there is tension too.) ∴ Particle does not slide back down the plane.
	Compares 'their' (sum of) friction (and tension) with the component of weight parallel to the plane.	AO2.4	R1	
	Determines that particle does not slide back down plane.	AO2.2a	R1	
				Total 10 marks

Q4.

(a) Work done = $\int_0^e \frac{\lambda x}{l}\,\mathrm{d}x$

SC1 $\int_0^e \frac{\lambda e}{l}\,de$

M1

$= \left[\frac{\lambda x^2}{2l}\right]_0^e$

SC1 $\int \frac{\lambda x}{l}\,dx$ *with no limits*

A1

$= \frac{\lambda e^2}{2l}$

A1

3

(b) (i) Using $T = \frac{\lambda x}{l}$:

$5g = \frac{392x}{1.6}$

$x = \frac{5g \times 1.6}{392}$

= 0.2

M1

Extension is 0.2 m

A1

2

(ii) When extension is 0.6 m, EPE = $\frac{\lambda x^2}{2l}$

B1 for 0.6

B1

$= \frac{392 \times (0.6)^2}{2 \times 1.6}$

M1

= 44.1 J

A1

3

(iii) Let y metres be distance particle is above A.

C of energy, when particle has speed 0.8 m s^{-1}, gives

$5 \times g \times y + \frac{392 \times (0.6 - y)^2}{2 \times 1.6} + \frac{1}{2} \times 5 \times 0.8^2$

M1 4 terms, 2 correct

M1A1 4 terms, 3 correct

M1A1

$= \frac{392 \times (0.6)^2}{2 \times 1.6}$

M1A2 4 terms correct

Ft answer to (b)(ii)

A1F

$49y + 122.5(0.6 - y)^2 + 1.6 = 122.5 \times 0.6^2$

$49y - 147y + 122.5y^2 + 1.6 = 0$

$122.5y^2 - 98y + 1.6 = 0$

$y = \frac{98 \pm \sqrt{98^2 - 4 \times 122.5 \times 1.6}}{2 \times 122.5}$

$$y = \frac{98 \pm 93.9148}{245}$$

= 0.016674 and 0.7833

if x used instead of 0.6 – y, A1 here for x = 0.5833...

A1

Speed first becomes 0.8 when y = 0.0167

E1

5

[13]

Q5.

(a) $T_{AC} = \frac{45}{0.4}(0.6 + x - 0.4) = 22.5 + 112.5x)$

M1: Two tensions in terms of x with ± a constant term.

A1: First correct tension.

M1A1

$T_{BC} = \frac{45}{0.4}(0.6 - x - 0.4) = 22.5 + 112.5x)$

A1: second correct tension

A1

Resultant = $T_{BC} - T_{AC}$

= 22.5 – 112.5x – (22.5 + 112.5x)

= –225x

A1: Tensions summed correctly.

A1

AG

4

(b) $9\frac{d^2x}{dt^2} = -225x$

$\frac{d^2x}{dt^2} = -\frac{225}{9}x = -25x$

M1: Showing second derivative = –25x.

M1

∴ SHM

A1: Correct deduction about SHM.

A1

2

(c) $\omega = 5$

M1: Using ω from part (b).

M1

Period = $\frac{2\pi}{5}$

A1: Correct period.
Accept AWRT 1.26.

A1

2

(d) $a = 0.1$

B1: Use of a = 0.1

B1

$v^2 = 5^2\,(0.1^2 - 0.05^2)$

M1: Equation with -0.05^2 or $0.05^2 -$
A1: Correct equation.

M1A1

$v = 0.433$ m s^{-1}

A1: Correct speed.

A1

4

(e) $x = 0.1\cos(5t)$

B1F: Seeing a = 0.1 of their earlier value.

B1F

B1F: Correct expression using their values.

B1F

2

[14]

Q6.

(a) $75\,\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = 75\,g - \frac{450}{12}\,x - 15\,\frac{\mathrm{d}x}{\mathrm{d}t}$

M1: equation with four terms of the correct format.
A1: Correct terms.

M1A1

A1: Correct signs.

A1

$75\frac{d^2x}{dt^2} = 750 - 37.5x - 15\frac{dx}{dt}$

$10\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 100$

A1: Correct result from correct working.

A1

AG

4

(b) CF

$10\lambda^2 + 2\lambda + 5 = 0$

M1: Correct auxiliary equation.

M1

$\lambda = \frac{-2 \pm \sqrt{2^2 - 4\times5\times10}}{2\times10} = -0.1 \pm 0.7i$

M1: Correct complex solutions.

M1

$x = e^{-0.1t}(A\sin(0.7t) + B\cos(0.7t))$

A1: Correct CF

A1

PI

$x = 20$

B1: Correct PI.

B1

$x = e^{-0.1t}(A\sin(0.7t) + B\cos(0.7t)) + 20$

dM1: Using initial conditions to find B.

dM1

$x = 0,\ t = 0 \Rightarrow B = -20$

A1: Correct value for B.

A1

$x = e^{-0.1t}(A\sin(0.7t) - 20\cos(0.7t)) + 20$

$\dot{x} = -0.1e^{-0.1t}(A\sin(0.7t - 14\cos(0.7t)) +$

$e^{-0.1t}(0.7A\cos(0.7t - 14\sin(0.7t))$

dM1: Correct $\dot{x}$

dM1

$\dot{x} = 12.5,\ t = 0 \Rightarrow A = 15$

dM1: Using initial conditions to find A.

dM1

$x = e^{-0.1t}(15\sin(0.7t) - 20\cos(0.7t)) + 20$

A1: Correct value for A and correct final expression.

A1

10

(c) $v = e^{-0.1t}(12.5\cos(0.7t) + 12.5\sin(0.7t))$

$v = 0$

M1: Setting derivative equal to zero.
A1: Correct equation including correct derivative.

M1A1

$\tan(0.7t) = -1$

dM1: Value for tan(0.7t).

dM1

A1: Correct value for tan(0.7t).

A1

$t = \frac{15\pi}{14} = 3.37$ s

A1: Correct time.

A1

5

[19]

Q7.

(a) using EPE = $\frac{\lambda x^2}{2l}$,

M1

EPE = $\frac{32 \times 2.2^2}{2 \times 0.8}$

B1 for 2.2

B1

= 96.8 J

A1

3

(b) by C of Energy, when next at rest, EPE (initial) = work done against friction + EPE (when at rest)

M1A1 for work done by friction or 5F

M1A1

$96.8 = F \times 5 + \frac{32 \times 1.2^2}{2 \times 0.8}$

M1 3 terms; A1 all correct

M1A1

$5F = 96.8 – 28.8$

B1 28.8

B1

frictional force is 13.6N

A1

6

(c) at B, tension is $\frac{32 \times 1.2}{0.8}$

= 48N

B1

tension > friction
hence particle starts to move

E1

2

(d) when particle is next at rest,
work done against friction is EPE at B

13.6 × distance = 28.8

M1

distance is 2.1176

= 2.12 m

CAO

A1

2

(e) total distance is 5 + 2.1176

ft from M1 in (d)
or total distance × 13.6 = original EPE, 96.8

B1

1

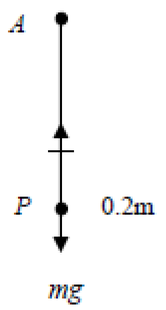
= 7.12 m

total distance is 7.12 m

[14]

Q8.

(a)



$T = 0.4g$

$T = 0.2k$

M1

$0.2k = 0.4g$

A1

$k = 2g$

$k = 19.6\text{Nm}^{-1}$

A1F

Both, accept use of λ and l

3

(b) extension = $0.2 + x$

B1

(i) $mg - T = m\ddot{x}$

M1

$0.4g - 19.6(x + 0.2) = 0.4\ddot{x}$

ft stiffness

A1F

$0.4\ddot{x} = -19.6x$

$\ddot{x} = -49x$ $(h = -49)$

A1F

4

ft stiffness

(ii) $\ddot{x} = -\omega^2 x$ *SHM*

(constant < 0)

ft stiffness provided h < 0

B1F

1

(iii) $T = \frac{2\pi}{\omega}$

$T = \frac{2\pi}{7}$

M1

T = 0.898 sec

AG

A1

2

(iv) max $v = a\omega$

= 0.1 × 7

M1

= 0.7 ms^{-1}

ft stiffness provided h < 0

A1F

2

[12]

Q9.

(a) Work done = $\int_0^e \frac{\lambda x}{l} \mathrm{d}x$

Condone lack of limits and 'dx'

M1

$= \left[\frac{\lambda x^2}{2l}\right]_0^e$

Must include limits from integral

A1

$= \frac{\lambda e^2}{2l}$

AG

A1

3

(b) (i) Using $T = \frac{\lambda x}{l}$, $7g = \frac{196x}{2}$

M1: could use 3g or 4g – at least 1 side correct

M1

$x = \frac{14g}{196}$

A1

= 0.7

A1

3

(ii) By C of Energy, when next at rest, EPE (initial) = PE change (for platform) + EPE (when at rest)

$$\frac{196 \times 0.7^2}{2 \times 2} = 4 \times g \times (0.7 - x) + \frac{196x^2}{2 \times 2}$$

M1: 3 terms (not including $\frac{1}{2}$ mv^2)
A1: 2 of 3 terms correct

M1A1

A1: all correct

A1

$2.45 = 2.8 - 4x + 5x^2$
$100x^2 - 80x + 7 = 0$

m1

$(10x - 7)(10x - 1) = 0$

A1

$x = 0.1$

[last A1: must give 0.1, not 0.1 and 0.7]

A1

6

Alternative

$$\frac{196 \times 0.7^2}{2 \times 2} = 4gX + \frac{196(0.7 - X)^2}{2 \times 2}$$

(M1)

(where X is distance moved upwards)

(A1)
(A1)

$4gX = 98 \times 0.7X + 49X^2$

(m1)

$X = 0,\ 0.6$

(A1A1)

(iii) Max speed when $T = mg$

M1

$4g = \frac{196x}{2}$

A1

$x = 0.4$

Or mid-point of values 0.2 and 0.6 above SC2

A1

3

[15]

Q10.

As particle moves, $T = \frac{mv^2}{r}$

M1

If radius is r, extension is $r - 1.2$

B1

Using $T = \frac{\lambda x}{l}$:

$T = \frac{192(r - 1.2)}{1.2}$

M1

$= 160(r - 1.2)$

A1

$$T = \frac{mv^2}{r} \Rightarrow 160(r - 1.2) = \frac{8 \times 3^2}{r}$$

M1

$160r^2 - 192r = 72$
(or $192r^2 - 230.4r = 86.4$)

A1

$20r^2 - 24r - 9 = 0$
$(10r + 3)(2r - 3) = 0$

M1

$r = 1.5$ or -0.3
Radius is 1.5

A1

or
using unknown as extension:
If extension is x, radius is 1.2 + x B1

Using $T = \frac{\lambda x}{l}$:

$T = \frac{192x}{1.2}$ *M1*

$= 160x$ *A1*

$T = \frac{mv^2}{r} \Rightarrow 160x = \frac{8 \times 3^2}{1.2 + x}$ *M1M1*

$192x + 160x^2 = 72$ *A1*
$20x^2 + 24x - 9 = 0$
$(10x - 3)(2x + 3) = 0$ *M1*
$x = 0.3$ *or* -1.5
Radius is 1.5 *A1*

[8]

Q11.

(a) When acceleration is zero, tension = gravitational force

$$\frac{784x}{16} = 80g$$

Both terms correct

M1

$x = 16, \; x + 16 = 32$ m

A1 for x = 16

A1

Length of cord is 32 m

A1

3

(b) (i) When bungee jumper comes to rest,

EPE = $\frac{784 \times x^2}{2 \times 16}$

M1

$= \frac{49x^2}{2}$

Change in PE = $80 \times g \times (16 + x)$

Or $80 \times g \times 65 - (80g[16+ x])$

(or $80g(49 - x)$)

M1

$\frac{49x^2}{2} = 80 \times 9.8 \times (16 + x)$

A1

$x^2 = 32x + 512$

$x^2 - 32x - 512 = 0$

AG

A1

4

(ii) $x = \frac{32 \pm \sqrt{32^2 + 2048}}{2}$

M1

$x = 43.7128$

A1

Distance below point of jump is 43.7 + 16 = 59.7 m

Distance between jumper and ground is 65 – 59.7

M1

= 5.29 m

Accept 5.287, 5.3

A1

4

[11]

Q12.

(a) Work done = $\int_0^e \frac{\lambda x}{l}\,dx$

M1

$= \left[\frac{\lambda x^2}{2l}\right]_0^e$

Needs limit of 0

A1

$= \frac{\lambda e^2}{2l}$

A1

Or
area under a straight line
= average force × distance
$= \frac{\lambda e^2}{2l}$

3

(b) (i) Using $T = \frac{\lambda x}{l}$

$5g = \frac{150 \times x}{0.6}$

M1

Extension is 0.196 m

A1

2

(ii) EPE = $\frac{\lambda x^2}{2l}$

$= \frac{150 \times (1.4)^2}{2 \times 0.6}$

M1

= 245 J

A1

2

(iii) When at point O,
EPE = 0
PE [relative to P] = 5 × g × 2

M1

KE [at O] = EPE [at P] – gain in PE [at O]

M1

$$\frac{1}{2}mv^2 = 245 - 10g$$

$$\frac{1}{2}.5.v^2 = 147$$

A1

$v^2 = 58.8$ Speed is 7.67 m s^{-1}

A1

4

[11]

Q13.

(a) $2g = \dfrac{49 \times x}{0.5}$

M1A1

$\underline{x = 0.2}$

A1

3

(b) EPE = $\dfrac{49 \times (0.2)^2}{2 \times 0.5}$

M1

= 1.96 (J)

A1

2

(c) (i) $1.96 = \dfrac{49 \times x^2}{2 \times 0.5} + 0.8 \times 9.8 \times (0.2 + x)$

All terms attempted for M1

M1

-1 EE from A3

A3

$x^2 + 0.16x - 0.008 = 0$

A1

5

(ii) $x = \frac{0.16 \pm \sqrt{0.16^2 + 4 \times 0.008}}{2}$

M1

$x = 0.04$

$x = 0.04$ only identified

A1

2

[12]

Q14.

(a) KE is loss in PE

$= 4 \times g \times 1.5$

M1 for $mgh = 58.8$ and then find v without finding KE

M1

$= 6g$ J or 58.8 J

A1

2

(b) When 3.5 m below O, extension is 2 m

EPE is $\frac{\lambda x^2}{2l} = \frac{\lambda(2)^2}{2 \times 1.5} = \frac{4}{3}\lambda$

M1

Change in potential energy of the particle is $4 \times g \times 3.5$

M1

$= 14g$ or 137.2

A1

$\frac{4}{3}\lambda = 14g$

$\lambda = 102.9$ N or 103 N

AG

A1

4

(c) When particle is 2.7 m below O,

EPE is $\frac{\lambda x^2}{2l} = \frac{\lambda(1.2)^2}{2 \times 1.5} = 49.392$

Accept 49.44 [from 103]

M1A1

Change in potential energy of the particle [from initial position] is
$4 \times g \times 2.7 = 10.8g$ or 105.84

B1

Conservation of energy:

$105.84 = \frac{1}{2} \times m \times v^2 + 49.392$

M1 for 3 terms and $4 \times g \times h$

M1

$2v^2 = 56.448$
Speed is $5.3126 \text{ m s}^{-1} = 5.31 \text{ m s}^{-1}$

CAO

A1

5

[11]

Q15.

(a) $\frac{100}{0.4}e = 10 \times 9.8$

Use of Hookes law and equilibrium

M1

$e = 0.392$ m

Correct length

A1

2

(b) $EPE = \frac{1}{2} \times \frac{100}{0.4} \times 0.6^2 = 45$ J

Use of EPE formula

M1

AG

Correct value from correct working

A1

2

(c) (i) $45 = \frac{1}{2} \times \frac{100}{0.4}(x - 0.4)^2 + \frac{1}{2} \times 10v^2 + 10 \times 9.8(1 - x)$

Expression for EPE with $(x \pm 0.4)^2$

M1

Correct EPE

A1

Four term energy equation

M1

$45 = 125(x - 0.4)^2 + 5v^2 + 98(1 - x)$

Correct GPE

B1

Correct equation

A1

$5v^2 = 98x - 98 + 45 - 125x^2 + 100x - 20$

Solving for v^2

dM1

$v^2 = 39.6x - 25x^2 - 14.6$ AG

Correct result from correct working

A1

7

(ii) $39.6x - 25x^2 - 14.6 = 0$

$25x^2 - 39.6x + 14.6 = 0$

$x = \dfrac{39.6 \pm \sqrt{39.6^2 - 4 \times 25 \times 14.6}}{2 \times 25}$

Solving quadratic

M1

= l or 0.584

Correct solutions

A1

$x = 0.584$

Appropriate value selected

SC Only correct answers given, award M1A1.

A1

3

[14]

Q16.

(a) $12.5 = \lambda \times \frac{0.1}{0.4}$

M1: subs. A1:All correct

M1A1

$\lambda = 50$

A1

3

(b) EPE = $\frac{50 \times (0.1)^2}{2 \times 0.4}$

Subs.

M1

= 0.625 J

PI A1: all correct

A1

$0.625 = \frac{1}{2} \times 0.2 \times v^2$

M1 use of principle

M1

ft EPE

A1F

v = 2.5 ms^{-1}

ft EPE

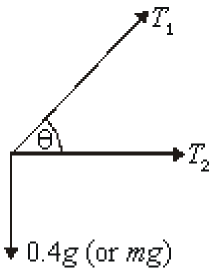
A1F

5

[8]

Q17.

(a)



Three forces evident;
T_1, T_2 clear – on diagram or in calculations

B1

1

(b) Vertically $T_1 \sin\theta = 0.4g$

Resolve vertically – component evident

$T \sin\theta = mg$ seen

M1 A1

$T_1 \left(\frac{4}{5}\right) = 0.4g$

M1

$T_1 = 0.5g = 4.9\text{N}$

AG obtained

A1

3

(c) $T_2 = 0.1k$

B1

1

(d) $a = r\omega^2 = 0.3 \times 5^2$

Calculation of a seen

B1

Force = $T_2 + T_1 \cos\theta$

(Both terms of horizontal resultant force attempted)

M1

$= 0.1k + 4.9 \times \frac{3}{5}$

Previous expression / result substituted

A1

$F = ma$

Their 'a' and 'F' (Dependent on first M1)

m1

$k = 0.6$

AG

A1

5

(e) EPE = $\frac{kx^2}{2} = \frac{0.6(0.1)^2}{2}$

$\frac{kx^2}{2}$ *seen and attempt to use*

M1

= 0.003 J

A1

2

[12]

Q18.

(a) Rope taut (momentarily) after 14m,

so PE = KE

$70g(14) = \frac{1}{2}(70)v^2$

attempt at conservation of energy
or $v^2 = u^2 + 2as$

M1

$v^2 = 28g$

$v \approx 16.6$

AWRT

A1

2

(b) (i) Initial PE = EPE at end

$70g(14 + x) = \frac{1}{2}(196)x^2$

attempt at conservation of energy equation
fully correct
For EPE (any stage)

M1 A1 B1

$70(14 + x) = 10x^2$

$7(14 + x) = x^2$

$0 = x^2 - 7x - 98$

Correct quadratic – AG

A1

4

(ii) $0 = (x - 14)(x + 7)$

any valid method for solving quadratic

M1

x = 14 or –7

$\therefore$ x = 14 m

A1

2

(iii) Hooke's law: $T = kx$

$\therefore$ T = 196(14)

Hooke's Law used; ft their x

B1f.t.

Newton's law: 70g – T = 70a

M1 attempt F = ma

M1A1

$$\therefore \frac{70(9.8) - 196(14)}{70} = a$$

–29.4 ms^{-2}

accept + or –

A1

4

[12]

(a) Rope taut (momentarily) after 14m,

so PE = KE

$70g(14) = \frac{1}{2}(70)v^2$

attempt at conservation of energy

or $v^2 = u^2 + 2as$

M1

v^2 = 28g

v ≈16.6

AWRT

A1

2

(b) (i) Initial PE = EPE at end

$70g(14 + x) = \frac{1}{2}(196)x^2$

attempt at conservation of energy equation

fully correct

For EPE (any stage)

M1 A1 B1

$70(14 + x) = 10x^2$

$7(14 + x) = x^2$

$0 = x^2 - 7x - 98$

Correct quadratic – AG

A1

4

(ii) $0 = (x - 14)(x + 7)$

any valid method for solving quadratic

M1

$x = 14$ or -7

$\therefore\ x = 14$ m

A1

2

(iii) Hooke's law: $T = kx$

$\therefore\ T = 196(14)$

Hooke's Law used; ft their x

B1f.t.

Newton's law: $70g - T = 70a$

M1 attempt F = ma

M1A1

$$\therefore\ \frac{70(9.8) - 196(14)}{70} = a$$

-29.4 ms^{-2}

accept + or –

A1

4

[12]

Q19.

(a) $2 \times 9.8 = \frac{\lambda}{0.5} \times 0.2$

Equilibrium considered to form equation in λ

M1

$\lambda = \frac{9.8}{0.2} = 49$ N

Correct equation

A1

Correct λ

A1

3

(b) $T = \frac{49x}{0.5} + 2g$

Equation for tension with two terms

M1

Correct equation

A1

$2\frac{d^2x}{dt^2} = 2g - (98x + 2g)$

Use of $F = m\frac{d^2x}{dt^2}$

M1

$\frac{d^2x}{dt^2} = -\frac{98}{2}x = -49x$

Correct equation

A1

ag *Correct result from correct working*

A1

5

(c) Period $= \frac{2\pi}{\sqrt{49}}$

Finding period

M1

Correct period

A1

$t = \frac{1}{4} \times \frac{2\pi}{7} = \frac{\pi}{14} = 0.224$ seconds

Dividing period by 4

M1

Correct time

A1

4

[12]

Q20.

(a) $mg = \frac{\lambda}{l} \times \frac{l}{4}$

Consideration of forces in equilibrium

M1

$\lambda = 4mg$

ag *Correct λ from correct working*

A1

2

(b) (i) $m\frac{d^2x}{dt^2} = mg - T$

$= mg - \frac{4mg}{l}\left(x + \frac{l}{4}\right)$

Two term expression for T

M1

Correct expression for T

A1

Three term equation of motion

M1

$= -\frac{4mg}{l}x$

Simplifying

m1

$\frac{d^2x}{dt^2} = -\frac{4g}{l}x$

ag *Correct result from correct working*

A1

5

(ii) Period $= 2\pi\sqrt{\frac{1}{4g}} = \pi\sqrt{\frac{1}{g}}$

Identifying ω

M1

Correct period

A1

2

[9]

Q21.

(a) $20 \times 9.8 = \frac{0.7\lambda}{2}$

Use of T = mg

M1 A1

$\lambda = \frac{2 \times 20 \times 9.8}{0.7} = 560$

Correct result from working

A1

3

(b) (i) $20 \times 9.8L = \frac{560(L-2)^2}{2 \times 2}$

Two term energy equation

Correct terms

Correct signs

M1 A1 A1

$196L = 140L^2 - 560L + 560$

Expanding and simplifying

m1

$5L^2 - 27L + 20 = 0$

Correct result from correct working

A1

5

(ii) $L = \frac{27 \pm \sqrt{27^2 - 4 \times 5 \times 20}}{2 \times 5}$

Solving a quadratic

M1

$= 4.51$ or 0.886

Correct solutions

A1

$L = 4.51$

Selecting the appropriate solution

A1

3

[11]

Q22.

(a) $T_{AP} = \frac{\lambda.3a}{4a} = \frac{3}{4}\lambda$

$T_{PB} = 4mg.\frac{a}{2a}$ = 2mg

Either

B1

Using $F = ma$ vertically

M1

$mg + T_{PB} = T_{AP}$

$\therefore\ mg + 2mg = \frac{3}{4}\lambda$

A1

$\lambda = 4mg$

A1

4

(b) (i) When particle is moved a distance x below the equilibrium position, forces acting on it are

mg, $T_{AP} = \frac{\lambda(3a+x)}{4a} = \frac{mg(3a+x)}{a}$,

M1

$T_{PB} = 4mg.\frac{(a-x)}{2a} = \frac{2mg}{a}(a-x)$

and resistance $\frac{1}{3}mk\dot{x}$

All four forces

M1

[forces 2 and 4 are upwards]

Using $F = ma$ vertically downwards

$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mk\dot{x}$

Dependent on both M1 above

m1

$m\ddot{x} = mg + \frac{2mg}{a}(a-x) - \frac{mg(3a+x)}{a} - \frac{1}{5}mk\dot{x}$

A1

$\ddot{x} - g - \frac{2g}{a}(a-x) + \frac{g(3a+x)}{a} + \frac{1}{5}k\dot{x} = 0$

$\ddot{x} + \frac{1}{5}k\dot{x} + \frac{3gx}{a} = 0$

A1

$10\frac{d^2x}{dt^2} + 2k\frac{dx}{dt} + 5k^2x = 0$

A1

6

(ii) Substituting $x = Ae^{nt}$,

$10n^2 + 2kn + 5k^2 = 0$

$n = \frac{-2k \pm \sqrt{4k^2 - 200k^2}}{20}$

M1

$= \frac{1}{10}(-k \pm 7k\mathrm{i})$

A1

$x = e^{-\frac{k}{10}t}(A\cos\frac{7}{10}kt + B\sin\frac{7}{10}kt)$

M1 A1ft

When $t = 0$, $x = \frac{a}{2}$, $A = \frac{a}{2}$

B1

Differentiating

$\frac{dx}{dt} = -\frac{k}{10}e^{-\frac{k}{10}t}(A\cos\frac{7}{10}kt + B\sin\frac{7}{10}kt) +$

$e^{-\frac{k}{10}t}\left(-\frac{7}{10}kA\sin\frac{7}{10}kt + \frac{7}{10}kB\cos\frac{7}{10}kt\right)$

M1 A1ft

When $t = 0$, $\frac{dx}{dt} = 0$, $0 = \frac{k}{10}A + \frac{7}{10}kB$

M1

$B = \frac{a}{14}$

$x = \frac{a}{14}e^{-\frac{k}{10}t}\left(7\cos\frac{7}{10}kt + \sin\frac{7}{10}kt\right)$

A1

9

[19]

Q23.

(a) $mg = \frac{\lambda e}{1}$

Equation for equilibrium

M1

$e = \frac{mgl}{\lambda}$

Correct extension

A1

2

(b) $T = -\frac{\lambda}{1}\left(x + \frac{mgl}{\lambda}\right)$

Expression for tension
Correct expression

A1

$m\frac{d^2x}{dt^2} = mg - T$

Application of Newton's 2nd law with T and mg

m1

$= mg - \frac{\lambda}{1}\left(x + \frac{mgl}{\lambda}\right)$

Simplified to correct final answer

A1

5

(c) $\omega = 2\pi\sqrt{\frac{\lambda}{ml}}$

$P = 2\pi\sqrt{\frac{ml}{\lambda}}$

Finding period

M1

Correct final answer

A1

2

(d) Increased by a factor of 2

Increased

B1

Factor of 2

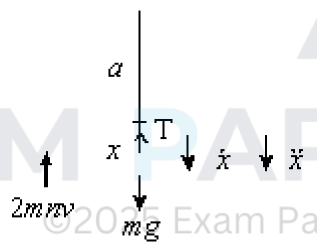
B1

2

[11]

Q24.

(a)



Using $F = ma$,

$m\ddot{x} = mg - \text{T} - 2mnv$

M1

$= mg - 6mn^2 a, \frac{x}{a} - 2mnv$

A1

$m\ddot{x} + 6mn^2 x + 2mnx = mg$

i.e. $\frac{d^2x}{dt^2} + 2n\frac{dx}{dt} + 6n^2 x = g$

A1

(b) Substituting $x = Ae^{pt}$

M1

$p^2 + 2np + 6n^2 = 0$

$p = \dfrac{\left(-2n \pm \sqrt{4n^2 - 24n^2}\right)}{2}$

$= -n \pm \sqrt{5}\, i\, n$

A1

$\therefore$ CF is $x = e^{-nt}$ (A cos $\sqrt{5}\, nt + \text{B} \sin \sqrt{5}\, nt$)

A1

PI is $x = \dfrac{g}{6n^2}$

B1

$\therefore$ GS is $x = e^{-nt}$ (A cos $\sqrt{5}\, nt + \text{B} \sin \sqrt{5}\, nt$

$= \dfrac{g}{6n^2}$

A1

When $t = 0$, $x = 0 \Rightarrow 0 = \text{A} + \dfrac{g}{6n^2}$

M1

$\therefore$ A = $-\dfrac{g}{6n^2}$

A1

When $t = 0$, $\dfrac{dx}{dt} = 0;$

$\dfrac{dx}{dt} = -ne^{-nt}(\text{A} \cos \sqrt{5}nt \times \text{B} \sin \sqrt{5}nt)$

$+ e^{-nt}\left(-\sqrt{5}n\, \text{A} \sin \sqrt{5}nt + \sqrt{5}n\, \text{B} \cos \sqrt{5}nt\right)$

$+e^{-nt}\left(-\sqrt{5}n \;\; A \sin \sqrt{5}nt + \sqrt{5}n \;\; B \;\; \cos \sqrt{5}nt\right)$

M1

$\Rightarrow 0 = -n\text{ A} \times \sqrt{5}n\text{B})$

$\therefore \text{ B} = \frac{1}{\sqrt{5}}\text{A} = -\frac{g}{6\sqrt{5}n^2}$

A1

$\therefore \; x = \frac{g}{6n^2} -$

$\left(\frac{g}{6n^2}\cos\sqrt{5}nt + \frac{g}{6\sqrt{5}n^2}\sin\sqrt{5}nt\right)e^{-nt}$

A1 10

(c) $\frac{dx}{dt} =$

$ne^{-nt} \left(\frac{g}{6n^2}\cos\sqrt{5}nt + \frac{g}{6\sqrt{5}n^2}\sin\sqrt{5}nt\right) -$

$e^{-nt}\left(-\frac{\sqrt{5}g}{6n}\sin\sqrt{5}nt + \frac{g}{6n}\cos\sqrt{5}nt\right)$

M1

$\frac{dx}{dt} = 0 \text{ when } \cos\sqrt{5}nt + \frac{1}{\sqrt{5}}\sin\sqrt{5}nt +$

$\sqrt{5}\sin\sqrt{5}nt\cos\sqrt{5}nt = 0$

$\sin\sqrt{5}nt = 0$

M1A1

First time at rest is when $t = \frac{\pi}{\sqrt{5}n}$

When $t = \frac{\pi}{\sqrt{5}n}$, $\frac{dx}{dt} = 0$ SC2

A1 4

[17]

Examiner reports

Q1.

In this question, students were able to demonstrate their understanding of principles of energy shown by a variety of correct approaches in part (b). The best approaches had clearly labelled diagrams which helped students obtain correct and consistent expressions for different types of energies. The mean mark for part (a) was 51%, the mean mark for (b) was 43% and the mean mark for (c) was 19%.

Part (a) was meant to be a simple application of conservation of energy when the cord reached its natural length. Many students applied the formulae correctly but did not give the final answer to two significant figures, thus losing the final mark. Required accuracy should match the value of g given in the question. A less common error was to use the height of the bridge above the river rather than the 25 m required.

There were many different approaches to part (b) and if students had used diagrams they would have been less confused and obtained more marks. The approaches seen were:

- calculation of the initial total energy (36750 J) and to compare this with the energy required for Dominic to reach the river (40000 J), deducing that there was insufficient energy to do so
- forming an energy equation and attempting to find the speed of Dominic at the surface of the river, deducing that he did not reach the river as v^2 was negative
- using x as the maximum extension, forming an energy equation to deduce that x = 23.6 m, deducing that Dominic would stop 1.4 m above the river and not get wet
- using x as the maximum length, forming an energy equation to deduce that x = 48.6 m, deducing that Dominic would stop 1.4 m above the river and not get wet
- using x as the minimum distance above the river, forming an energy equation to deduce that x = 1.4 m deducing that Dominic would not get wet.

All these methods proved to be successful for some students. However, without clear diagrams other students often got confused about what they were finding, leading to incorrect combinations of the different types of energies. Sometimes an x that had started out as a maximum length became an extension instead.

Part (c) linked to part (b) and required students to realise that if Dominic was not a particle then he would have height and hence could get wet if he was taller than 1.4 m. Any reference to air resistance here scored zero marks.

Q4.

Many students were unable to show convincingly that the work done in stretching the string was $\frac{\lambda e^2}{2l}$. An integration using a variable, usually x, was required. A number of students were penalised for using either $\int_0^e \frac{\lambda e}{l}\,de$ or $\int \frac{\lambda x}{l}\,dx$ (without including limits). Parts (b)(i) and (b)(ii) were usually answered correctly although a minority of students used the incorrect formulae for tension and elastic potential energy. In part (b)(iii), most students included terms in kinetic energy, gravitational potential energy and elastic potential energy but many did not include elastic potential energy at both A and the point when its speed was 0.8 ms^{-1}. Another common error was in using the same variable for the height that the particle moved (in the potential energy gained) and for the extension when the particle's speed was 0.8 ms^{-1}.

Q5.

In part (a), there were many good responses, but in a number of cases there were issues with the students not obtaining correct expressions for the tensions in the two strings. One common error was to simply give the tension as $T = \frac{45x}{0.4}$ and take no account of the fact that the string was already stretched. Another common error was to give the expressions of the form $T = \frac{45x}{0.4}(0.8 \pm x)$ for the tension. Most of these approaches "produced" the printed answer, but credit was only given where this was correctly obtained from correct working. Part (b) was usually done well although some students did not include the mass and ended up with an incorrect value for ω of 15 which they used later in the question. Part (c) followed on and was also done well in the main. In part (d) several students correctly used the formula $v^2\ \omega^2(a^2 - x)^2$, but a number of students used incorrect values for a. The values 0.5 and 0.7 were seen several times. In part (e), students often gave answers of the correct form, but where they had an incorrect amplitude or value for ω from earlier incorrect working this was carried through into their answers.

Q6.

Part (a) was done well by many students, but some struggled to provide a convincing justification for the differential equation. A clear statement of the resultant force and then a clear application of Newton's Second law would have helped some students.
The majority of students did very well with the solution of the differential equation and seemed to be comfortable with this part of the question. The most common error seen was probably to use $\dot{x} = 0$ rather than $\dot{x} = 12.5$. There were however a whole variety of minor arithmetical and algebraic errors. Part (c) was done well but obviously some students were using incorrect expressions for the speed.

Q7.

Part (a) of this question was answered successfully by most candidates. Although many candidates were successful in part (b), a few attempted to use kinetic energy. Relatively few candidates were successful in part (c), with most candidates stating that when the particle was at *B* the string was stretched and hence the string had elastic potential energy and thus the particle must move. The fact that the particle would only move again if the tension in the string was greater than the friction, found in part (b), was only considered by the better candidates. Those candidates who had answered part (b) correctly usually answered parts (d) and (e) correctly.

Q8.

This was another popular question for many, although a few failed to appreciate the vertical nature of the motion, falling down in part (b)(i). Parts (a) and (b)(ii) and (iii) were mostly successful but a few could not provide a correct formula for the maximum speed for this motion.

Q9.

Many candidates were unable to show convincingly that the work done in stretching the string was $\frac{\lambda e^2}{2l}$ An integration using a variable, usually x, was required.

Part (b)(i) was usually answered correctly. In part (b)(ii), most candidates appreciated that

they needed to consider gravitational potential energy and elastic potential energy. However, many candidates did not appreciate that the string was still stretched when the 4 kg block was next at rest. Often candidates used terms in kinetic energy, gravitational potential energy and elastic potential energy and then equated the speed to zero.
In part (b)(iii), the extension x when the speed is at a maximum was usually found by using $T = mg$. A few candidates used techniques involving the maximising of v^2 ; impressively, sometimes these were totally successful.

Q10.

Some candidates assumed that energy needed to be considered. Those who did equate tension to be $m\frac{v^2}{r}$ sometimes found $m\frac{v^2}{r} = \frac{\lambda x}{1.2}$ but then often failed to notice that $r = 1.2 + x$, which reduced their equation to one with only one unknown.

Q11.

Many candidates experienced difficulty with this question. In part (a), candidates rarely used the fact that the acceleration of the bungee jumper was zero when the tension in the cord was equal to the gravitational force. Candidates tried to use energy considerations in part (b)(i), but did not use the correct difference in height for potential energy. There was a significant amount of 'creative' algebra to obtain the printed equation. In part (b)(ii), most candidates solved the given equation quickly, and many then subtracted this from 65 to find the distance between the bungee jumper and the ground. The three significant figure solution to the quadratic resulted in a two significant figure answer which was not penalised. This rarely concerned candidates; only the better ones found the four significant figure solution to the quadratic in order to obtain a three significant figure answer.

Q12.

Part (a) tested work done = $\int F\,dx$. Few candidates found $\int_0^e \frac{\lambda x}{l}dx$ correctly. Instead of integrating, a few candidates used the value of the integral to be the area under the line $y = \frac{\lambda x}{l}$. Unfortunately, many candidates used techniques which were not credited: for example, elastic potential energy is $\frac{\lambda x^2}{2l}$ and x = e; or work done = maximum force × half the distance moved, without a clear justification.

Part (b)(i) and (ii) were completed well. In part (b)(iii), most candidates attempted to conserve kinetic energy, potential energy and elastic potential energy, but often the signs of the three terms in the equation $\frac{1}{2}mv^2 - 245 + 10g = 0$ were incorrect.

Q13.

Candidates were often very successful in parts (a) and (b). Most knew how to start part (c)(i) but elastic energy terms were sometimes omitted and there were many errors in potential energy terms. Part (c)(ii) was successful but not all selected the appropriate solution.

Q14.

In part (a), many candidates used conservation of energy, not to find the kinetic energy when the string became taut, but to find the velocity when the string became taut. Then they used KE = $\frac{1}{2}mv^2$ to find the kinetic energy when the string became taut. The simpler method of using conservation of energy with gain in kinetic energy equalling the loss in potential energy was rarely seen; this of course immediately gave the required kinetic energy. In part (b), most candidates used the change in potential energy to be the change in EPE and readily found λ.

Q15.

Parts (a) and (b) of this question were done very well, with many candidates gaining full marks on both parts. Part (c) (ii) was also done well, but some candidates did not show both solutions to the quadratic as part of their working. In questions like this candidates should be encouraged to show both solutions and then select the appropriate solution in the context under consideration. Part (c)(i) was very challenging and there were relatively few correct solutions. The three main problems were:

- Not including the elastic potential energy when released,
- Using an incorrect expression for the extension of the string when forming an expression for the elastic potential energy,
- Using an incorrect distance when forming an expression for the gravitational potential energy.

Q16.

Part (a) was done well, with candidates able to obtain the printed result convincingly. In part (b) there were many successful solutions. However, some candidates assumed the force on the particle to be constant and applied Newton's Second Law followed by the equations of constant acceleration.

Q17.

This proved to be a demanding question although a significant proportion of candidates solved it well. The force diagram in part (a) was rarely correct with equal tensions shown or, even worse, the tension in OP missing and sometimes a normal reaction appearing at P. Those candidates who had not clearly marked different tensions were able to recover the mark if their working out indicated the use of two different tensions.

In parts (b) and (d) many candidates struggled to set up the equations required. The answer in part (b) was arrived at through incorrect methods to match the answer given. In part (d) mrw^2 was often seen, but a number of candidates thought that this was directed along QP or outwards from P.

Parts (c) and (e) were more successful and it was pleasing to see the correct formulae in use.

Q18.

This was the most discriminating question on the paper. Part (a) did not particularly trouble any candidate.

In part (b), candidates who had been taught the $\frac{\lambda x^2}{2l}$ and the $\frac{\lambda x}{l}$ formulae were often not able to complete the question successfully. The specification is clear in its expectation. Weaker candidates often got muddled by mixing up energy terms from different points in the motion. It was pleasing to see that where a candidate had drawn clear diagrams the answer followed easily. It is worrying that many did not score two marks on the quadratic equation. Incomplete methods, invalid methods and failing to clearly rule out –7 as a solution were some of the reasons. Part (b)(iii) was scarcely seen correct, as many candidates did not realise that $mg - T = ma$ was required.

Q19.

The candidates did well with part (a) and for quite a number of candidates these were the only marks that they obtained on this question.

In part (b), the majority of the candidates totally ignored the weight of the sphere and produced invalid arguments. There were some candidates who could produce the required result without difficulty.

In the last part, there were quite a few candidates who found the period, but did not realise that the time needed was a quarter of the period.

Q20.

Part (a) of this question was done very well by almost all of the candidates. Part (b)(i) was very much more demanding and there were very few correct solutions. Most of the candidates did not find a two term expression for the tension in the string and did not include gravity when applying Newton's second law. In contrast, part (b)(ii) was done quite well, although some candidates did have difficulty simplifying their results.

Q21.

Part (a) was done well, with the printed answer helping some candidates. However, some candidates did try to use elastic potential energy in their solutions. In part (b), several candidates struggled to form a correct equation. Those who did normally went on to produce the required equation. Common errors were to omit the gravitational potential energy or to use $L - 2$ when finding the gravitational potential energy. Part (c) was often done well, but there were some candidates who seemed unaware of any of the techniques that could be used to solve a quadratic equation.

Q22.

Most candidates answered part (a) correctly but a number did not use $mg + T_{PB} = T_{AP}$ and fiddled the algebra to give the printed result, for which they gained no marks.

In part (b), candidates had to find tensions in the two springs AP and PB, with extensions $(3a + x)$ and $(a - x)$ respectively, and then use $F = ma$ downwards, giving $m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mkx$. Without these equations candidates were not given credit for producing the printed differential equation.

All candidates used the printed equation given in part (b)(i) to find x in part (ii). A number of candidates lost k in a variety of places, but virtually all obtained a result similar to that required.

Q23.

There were many good answers to part (a). Virtually all those who realised that they needed to consider when the forces were in equilibrium gained both marks. Part (b) was found to be very much more difficult. Several candidates did not include the weight of the particle at any stage of their solutions. Some others produced very contrived arguments to produce the negative sign in the differential equation. Part (c) was often done well, and those who did not gain marks on this question did not seem to know how to proceed. In part (d), several of the candidates stated that the period would increase, but a smaller number were able to quantify the increase correctly.

Q24.

Many candidates attempted to find the extension of the string in the equilibrium position, although in this question it was not relevant. These candidates had no mg force present when they found their differential equation. As this equation was printed on the question paper, candidates who then 'created' the mg force rarely scored marks in part (a). In part (b), candidates usually obtained the general solution $x = e^{-nt}(A\cos\sqrt{5}nt + B\sin\sqrt{5}nt)$, and most attempted to find the particular solution. They then showed the correct method for obtaining the actual solution. Those who correctly found the answer to part (b) often substituted $t = \frac{\pi}{\sqrt{5}n}$ to show that the particle was at rest.

Only the better candidates differentiated ther answer to show that the particle was at rest.